Week 3

Census Data Exploration

Course zoom link for Summer Quarter 2025

July 15, 2025

*Note that this course will be recorded🎥

Hands on Lab

- First, grab the course material, and "pull" it into your JupyterHub space:
- Git related
- Note that you have to do this at the start of every lecture to get the latest material.

Assignments (due 11:59 pm Monday, July 21st the day before class)

Project Assignment #2

Census Data Exploration

Step 1: Plan

For this project assignment, you will create and submit a Jupyter Notebook with 2-3 different census variables analyzed and visualized. The notebook should choose census indicators that are relevant to your research project. For example:

- you may choose to do a race profile for Los Angeles, and the other member may choose to do the same race profile for New York
- you may choose to do a transportation profile for Los Angeles, or you may choose an education profile also for Los Angeles
- you may choose a different time period to do a temporal analysis.
- **please see** the lab notebook on tips to download from the Census data portal

Additional census datasets to work with for assignment

- LA Geohub
- LA County GIS
- Living Atlas GIS data
- I have uploaded Census tracts here on the Course website for your use from 2024 Living Atlas link above. HINT: Refer to fields available from this Demographics data link.
- LA County Boundary
- Census data portal
- TIGER/Line 2024 census tracts. HINT: use this geospatial data source to join census data.
- EJScreen Table data
- EJScreen Table metadata

Warnings:

Do not copy and paste the class lab. Make sure to start a brand new notebook.

If you are working on a project outside of the United States, consult with me to find an alternative dataset to explore.

Step 2: Program

1. The first cell of the notebook should be a markdown cell that includes a title, description, and author of the assignment.
2. Download census variables relevant to your research question (if you are working outside of the United States, consult with me to find an alternative dataset)
3. Create a Jupyter Notebook and use pandas/geopandas to explore and visualize the data
4. Produce several charts, including one or more maps
5. Make sure to document each procedure with markdown cells with relevant headers
6. Run the cells in the notebook, and make sure the notebook "reads" from top to bottom, telling a story
7. Upload each Jupyter Notebook along with the associated census data file to your project repo

Step 3: Submit

- Submit your assignment by posting a link to your notebook here.